

11. Code-Layout, Readability and Reusability

Date	4 July 2026
Team ID	INDIVIDUAL
Project Name	AI-Powered Debt Relief & Financial Recovery Platform <u>FinRelief AI</u>
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces for Python, 2 for JS
2	Proper File Structure	Files and folders are logically organized	Yes	Separated backend/ and frontend/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., <u>monthly_surplus</u> , <u>loanData</u>
4	Function / Method Names	Functions are descriptively named	Yes	e.g., <u>generate_negotiation_letter()</u>
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings used in Python files
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate ai_engine.py and financial_engine.py

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6	Modular Design	Code is split into reusable functions/modules	Yes	Separate ai_engine.py and financial_engine.py
7	No Redundant Code	Duplicate or unused code is removed	Yes	DRY principle followed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-catch blocks in frontend and backend

Reusable Components/Modules:

S.No	Component/Module Name	Language / Technology Where Reused	Reusability Level (High/Medium/Low)
1	calculate_financial_health()	Python (Backend)	High (Used in dashboard and advisor)
2	generate_negotiation_letter()	Python + Gemini API	High (Used for all loan types)
3	Axios API Client	JavaScript (Frontend)	High (Used for all API calls)